

One-at-a-Time Perturbation Test

How the PLUS and MINUS one-at-a-time pass rates were generated — materials, procedure, decision rules and reproduction steps.

1. Purpose

The perturbation test asks whether the LLM sub-rubric scorer responds to the evidence a rubric criterion actually depends on. Two directions are tested. **PLUS** adds a sentence supplying evidence for one criterion and checks that the criterion's score rises. **MINUS** removes the passages supporting one criterion and checks that its score falls.

The original design edited a whole dimension at once: a single PLUS report received all eight sub-rubrics' sentences and a single MINUS report had all eight sets of passages removed. That design cannot attribute a score change to the targeted criterion, because seven unrelated edits were applied to the same document. The one-at-a-time design isolates a single edit per report, which makes attribution possible: a score change can be attributed to the edit that was made.

2. Materials and unit of analysis

The unit is a **cell**: one proposal × one dimension × one sub-rubric.

Component	Value
Proposals	12 (p1–p8, pA–pD)
Dimensions	strategy, objectives, advantages, team, feasibility
Sub-rubrics per dimension	8 (R1–R4 document-verifiable, R5–R8 judgment-dependent)
Total cells	$12 \times 5 \times 8 = 480$
Source reports	src/data/reports/<pid>__openai_final_report.md
Scorers	src/tools/generate_<dim>_rubric_scores.py (unmodified)
Score scale	1–5 integer per sub-rubric

Each scorer call returns all eight sub-rubric scores for one report, so a single call produces eight cell values.

3. Baseline

The unmodified report is scored first; those 480 scores are the reference against which every perturbation is judged. A cell's baseline is that specific sub-rubric's own score in the unmodified report — not a dimension average.

The baseline is scored a **second time** under identical settings. This run is never used for pass/fail; it exists only to quantify run-to-run noise. **444 of 480** sub-rubric scores (92.5%) were identical across the two runs, which characterises the scorer's run-to-run variability.

4. Controls applied to every scorer call

Baseline and perturbed variants are scored under identical conditions. The one-at-a-time scripts import `rescore_corrected.py` and reuse its scorer wrapper rather than reimplementing it, so no setting can drift between conditions.

- **Temperature 0.0** for every call, baseline and variant alike.
- **No AI expert-opinion JSON.** That artefact paraphrases the report's own commentary; supplying it to the baseline only would have reintroduced the very text a MINUS edit removes. `load_scores()` raises if a score file records an expert path.

- **Neutral report identifiers.** Every report is written under a SHA-1 derived id, so the `proposal_id` visible to the scorer reveals nothing about the variant:

```
digest = hashlib.sha1(f"corrected_v2|{pid}|{dim}|{variant}".encode()).hexdigest()[:10]
report_id = f"r2x{digest}__openai"          # e.g. r2xa3f91c2b08__openai
```

- **Single insertion point.** PLUS text is inserted immediately before the ##### Team and governance heading for every dimension, so the insertion position is not a free variable across conditions.

5. Building the perturbed reports

One report per cell. Each is a full copy of the original with exactly one criterion's evidence changed; everything else is byte-identical.

5.1 PLUS — add one sentence

The sentence aimed at the target sub-rubric is inserted at the marker. Sentences come from `added_text_review.py` where the originally generated sentence was found to be misaligned with its criterion, otherwise from the run's `rubric_coverage` content. All 480 cells receive a report.

```
def build_text(runner, original, sentence):
    text = runner.adaptive_inject_plus(original, sentence, rc.PLUS_MARKER)
    if text == original + "\n\n" + sentence:
        raise RuntimeError("PLUS marker not found")
    if text.find(sentence) + len(sentence) > rc.SCORER_WINDOW:
        raise RuntimeError("PLUS sentence is past the scorer window")
    return text
```

5.2 MINUS — replace one criterion's passages

Every occurrence of the target criterion's supporting passages is replaced by a fixed neutral sentence drawn from `removed_text_review.py`. Replacement rather than deletion keeps document length and structure roughly constant, so the scorer is not reacting to a shorter report. Only cells whose report contains something to remove receive a report: **170 of 480**.

```
# example — p1 / advantages / R5 Defensibility
removed : "The patent application for the heart failure large model indicates
          a commitment to innovation and intellectual property protection."
becomes  : "The proposal does not describe competitive barriers or
          differentiation from competitors."
```

5.3 Pre-flight validation

Both scripts support `--build_only`, which writes every report and runs all assertions without spending a single LLM call. The scorers truncate input at **12,000 characters**; this guard is the reason the corrected run supersedes the earlier adaptive run, in which injected PLUS text could fall outside the window and silently never reach the model.

- PLUS: the inserted sentence must end inside the 12,000-character window.
- MINUS: no removed passage may still appear in the first 12,000 characters.
- MINUS: every listed passage must be found in the source text, or the build fails.

6. Scoring

Reports are scored by the unmodified dimension scorer, six in parallel, skipping any report already scored so a failed run can be resumed. Any scorer error aborts the run rather than producing a partial table.

```
todo = [(dim, rid) for (pid, dim, key), rid in cells.items()
        if not rc.score_path(runner, dim, rid).exists()]
with ThreadPoolExecutor(max_workers=args.workers) as pool:
    errors = [... pool.map(lambda t: rc.run_scorer(runner, *t), todo) ...]
```

7. Decision rule

Each cell is judged by comparing the target sub-rubric's perturbed score with its own baseline. The rule is inherited unchanged from `aggregate_subrubric_passrates.py`.

Outcome	Condition
PASS (PLUS)	perturbed score > baseline
PASS (MINUS)	perturbed score < baseline
FAIL	the score did not move in the tested direction
EXCLUDED	PLUS at baseline 5, or MINUS at baseline 1 — the score cannot move
UNTESTABLE	MINUS only: the report contained no passage supporting the criterion

Pass rate = PASS / (PASS + FAIL). Excluded and untestable cells are reported in their own columns and are never counted as passes. Treating untestable cells as passes was a defect of the earlier run.

8. Coverage

Direction	Reports built	Tested	Excluded	Untestable	Total
PLUS	480	478	2 (baseline 5)	—	480
MINUS	170	127	43 (baseline 1)	310	480

MINUS coverage is limited by the source reports: 310 of 480 criteria had no supporting passage to remove. Eleven of the forty sub-rubrics are untestable in all twelve proposals and carry n/a rather than a rate.

9. Aggregation

Cell outcomes are tallied per sub-rubric, then rolled up by dimension × rubric type and finally across all dimensions. Every aggregate row is an exact sum of its component cells; rates are recomputed from the summed counts rather than averaged across rows.

10. Headline results

Group	PLUS one-at-a-time	MINUS one-at-a-time
All cells	300 / 478 = 0.628	115 / 127 = 0.906
Document-verifiable	172 / 238 = 0.723	65 / 67 = 0.970
Judgment-dependent	128 / 240 = 0.533	50 / 60 = 0.833

For comparison, the all-at-once design on the same cells yields PLUS 0.900 and MINUS 0.984. The difference arises because in those reports all eight criteria were edited simultaneously, so a target's movement cannot be attributed to the edit aimed at it.

11. Reproduction

```
python3 rescore_corrected.py --build_only # baseline + all-at-once, no LLM calls
python3 rescore_corrected.py # score; writes results.json
python3 rescore_minus_single.py --build_only # 170 single-removal reports
python3 rescore_minus_single.py # score; writes minus_single.json
python3 rescore_plus_single.py --build_only # 480 single-sentence reports
python3 rescore_plus_single.py # score; writes plus_single.json
python3 build_perturbation_review_xlsx.py # assemble the workbook
```

Artefact	Path
Baseline & all-at-once results	src/data/sensitivity_results/corrected_v2/results.json
MINUS one-at-a-time	src/data/sensitivity_results/corrected_v2/minus_single.json
PLUS one-at-a-time	src/data/sensitivity_results/corrected_v2/plus_single.json

Artefact	Path
Curated added text	UpdatedThesisWork/added_text_review.py
Curated removed text	UpdatedThesisWork/removed_text_review.py
Pass-rate workbook	UpdatedThesisWork/perturbation_review_one_at_a_time.xlsx

12. Limitations

- **MINUS coverage.** Only 127 of 480 cells are testable; the 0.906 rate rests on a quarter of the grid.
- **Headroom confounds PLUS.** Pass requires a rise, so pass rates fall steeply with baseline (0.78 at baseline 1 and 2, 0.35 at baseline 3, 0.23 at baseline 4). Where a criterion is already satisfied, a non-rise may be correct scorer behaviour rather than insensitivity.
- **Rubric-type labels are imperfect.** Several criteria labelled document-verifiable behave as holistic judgments (e.g. objectives R1 Problem Clarity, 0/12). Per-dimension type contrasts rest on roughly 48 cells and should not be read as significant.
- **Single scorer and provider.** All results come from one model at temperature 0; run-to-run variation is characterised, but cross-model variation is not.

Appendix A Per-sub-rubric results

Passed / tested, with the rate. Excluded and untestable counts are shown separately; each sub-rubric row covers all 12 proposals.

dimension	sub-rubric	type	PLUS	rate	exc	MINUS	rate	exc	untest
strategy	R1 Go-to-Market	doc	11/12	0.917	0	2/2	1.000	0	10
strategy	R2 Milestones	doc	10/12	0.833	0	3/3	1.000	3	6
strategy	R3 Revenue	doc	12/12	1.000	0	0/0	n/a	0	12
strategy	R4 Regulatory	doc	10/12	0.833	0	0/0	n/a	0	12
strategy	R5 Moat	judg	12/12	1.000	0	8/8	1.000	2	2
strategy	R6 Partners	judg	10/12	0.833	0	5/5	1.000	5	2
strategy	R7 Team Fit	judg	1/12	0.083	0	10/12	0.833	0	0
strategy	R8 Timing	judg	6/12	0.500	0	3/3	1.000	0	9
objectives	R1 Problem Clarity	doc	0/12	0.000	0	9/10	0.900	0	2
objectives	R2 Market Size	doc	12/12	1.000	0	0/0	n/a	0	12
objectives	R3 Buyer Pathway	doc	7/12	0.583	0	0/0	n/a	1	11
objectives	R4 Unmet Need Evidence	doc	10/12	0.833	0	1/1	1.000	0	11
objectives	R5 Why Now	judg	9/12	0.750	0	1/1	1.000	3	8
objectives	R6 Competitive Context	judg	5/12	0.417	0	3/3	1.000	1	8
objectives	R7 Problem-Sol Fit	judg	2/12	0.167	0	4/6	0.667	0	6
objectives	R8 Addressability	judg	4/12	0.333	0	0/0	n/a	2	10
advantages	R1 Mechanism Novelty	doc	3/12	0.250	0	10/10	1.000	0	2
advantages	R2 IP Status	doc	11/11	1.000	1	4/4	1.000	1	7
advantages	R3 Performance Proof	doc	10/12	0.833	0	2/2	1.000	4	6
advantages	R4 Competitor Benchmark	doc	5/12	0.417	0	0/0	n/a	0	12
advantages	R5 Defensibility	judg	7/12	0.583	0	3/3	1.000	1	8
advantages	R6 Platform Potential	judg	5/12	0.417	0	2/2	1.000	0	10
advantages	R7 Validation Signals	judg	9/12	0.750	0	1/1	1.000	2	9
advantages	R8 Adoption Readiness	judg	12/12	1.000	0	0/0	n/a	1	11
team	R1 Composition	doc	3/12	0.250	0	5/5	1.000	0	7
team	R2 Credentials	doc	4/12	0.333	0	11/11	1.000	0	1
team	R3 Track Record	doc	10/12	0.833	0	3/3	1.000	0	9
team	R4 Governance	doc	9/12	0.750	0	2/3	0.667	0	9
team	R5 Capacity	judg	9/12	0.750	0	0/1	0.000	0	11
team	R6 Complementarity	judg	0/12	0.000	0	5/8	0.625	0	4
team	R7 Cohesion	judg	11/12	0.917	0	0/0	n/a	0	12
team	R8 Key-Person Risk	judg	7/12	0.583	0	0/0	n/a	0	12
feasibility	R1 Budget Detail	doc	10/12	0.833	0	4/4	1.000	2	6
feasibility	R2 Funding Secured	doc	11/11	1.000	1	3/3	1.000	3	6
feasibility	R3 Infrastructure	doc	12/12	1.000	0	5/5	1.000	5	2
feasibility	R4 Technical Readiness	doc	12/12	1.000	0	1/1	1.000	3	8

dimension	sub-rubric	type	PLUS	rate	exc	MINUS	rate	exc	untest
feasibility	R5 Risk Mitigation	judg	2/12	0.167	0	3/5	0.600	1	6
feasibility	R6 Resource-Timeline Fit	judg	6/12	0.500	0	2/2	1.000	0	10
feasibility	R7 Financial Realism	judg	6/12	0.500	0	0/0	n/a	0	12
feasibility	R8 Sustainability	judg	5/12	0.417	0	0/0	n/a	3	9
strategy	— all —	all	72/96	0.750	0	31/33	0.939	10	53
strategy	— all —	doc	43/48	0.896	0	5/5	1.000	3	40
strategy	— all —	judg	29/48	0.604	0	26/28	0.929	7	13
objectives	— all —	all	49/96	0.510	0	18/21	0.857	7	68
objectives	— all —	doc	29/48	0.604	0	10/11	0.909	1	36
objectives	— all —	judg	20/48	0.417	0	8/10	0.800	6	32
advantages	— all —	all	62/95	0.653	1	22/22	1.000	9	65
advantages	— all —	doc	29/47	0.617	1	16/16	1.000	5	27
advantages	— all —	judg	33/48	0.688	0	6/6	1.000	4	38
team	— all —	all	53/96	0.552	0	26/31	0.839	0	65
team	— all —	doc	26/48	0.542	0	21/22	0.955	0	26
team	— all —	judg	27/48	0.562	0	5/9	0.556	0	39
feasibility	— all —	all	64/95	0.674	1	18/20	0.900	17	59
feasibility	— all —	doc	45/47	0.957	1	13/13	1.000	13	22
feasibility	— all —	judg	19/48	0.396	0	5/7	0.714	4	37
ALL	— all —	all	300/478	0.628	2	115/127	0.906	43	310
ALL	— all —	doc	172/238	0.723	2	65/67	0.970	22	151
ALL	— all —	judg	128/240	0.533	0	50/60	0.833	21	159